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REVIEW

Acyclovir dosing in herpes encephalitis: A scoping review

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ABSTRACT

Background: Herpes encephalitis, a rare yet potentially fatal viral infection, is treated exclusively with acyclovir, the sole antiviral medication used for this condition. Acyclovir recommended dose is 10 mg/kg/dose intravenous every 8 hours; however, it is unclear what body weight should be utilized in obese patients. Using the ideal body weight may result in subtherapeutic ineffective concentrations, while utilizing the actual body weight might result in acyclovir induced adverse effects, either nephrotoxicity or neurotoxicity or both.

Objective: The objective of this scoping review is to explore existing evidence regarding acyclovir dosing for obese patients afflicted with herpes encephalitis.

Methods: MEDLINE, EMBASE, Scopus, Web of Science, and CINAHL databases were searched on 26 May 2023, with no language restrictions. Two independent reviewers utilized the Covidence software to carry out the screening and selection of the articles. A total of 22 articles were included in the current review.

Results: The prevalence of acyclovir-associated nephrotoxicity ranged from 13% to 21%, while the prevalence of neurotoxicity was not clearly defined. However, there is lack of evidence regarding what may arise from subtherapeutic concentrations. An approach has been suggested to help clinicians to give the most appropriate acyclovir dose to herpes encephalitis patients. Patients with normal kidney function could receive the normal doses based on actual weight if normal weight and based on adjusted body weight if obese. On the other hand, if the patients are experiencing augmented renal clearance, they could receive up to the maximum recommended doses.

Conclusion: Overall, there is a lack of consistency on which body weight to use to calculate acyclovir dose in obese patients. So it is recommended that further studies compare the concentration of intravenous acyclovir between obese and nonobese patients and relating the resultant concentration with patient outcomes.

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Background

Herpes simplex virus (HSV) and herpes zoster encephalitis are uncommon conditions but potentially fatal viral infections affecting the central nervous system. Globally, the annual incidence of herpes simplex encephalitis is approximately 2 to 4 cases per 100,000 individuals. In the United States, HSV-1 encephalitis accounts for 10% to 20% of the 20,000 annual cases of viral encephalitis.[1,2] It is proven that acyclovir is the drug of choice to treat herpes encephalitis based on the guidelines with a level of recommendation A-I.[3] Acyclovir greatly decreases the morbidity and mortality of herpes encephalitis if it is started immediately.[4] If left untreated the mortality rate is 70% which could be reduced to 20% if acyclovir treatment was started as early as possible.[5,6]

Acyclovir is a nucleoside analog used as an antiviral drug that was discovered in 1974 by Schaeffer HJ et al.[7,8] It has antiviral activity against HSV and varicella zoster virus.[9] However, besides its clinical activity, in higher concentrations it may cause nephrotoxicity and/or neurotoxicity which require careful dose adjustment and concentration monitoring, if available. Furthermore, acyclovir neurotoxicity has been attributed to elevated concentrations of 9-(carboxymethoxymethyl) guanine (CMMG), an acyclovir metabolite.[10-12]

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Key Points

Background:

- Acyclovir is the drug used to treat herpes encephalitis, which is a potentially fatal disease with severe morbidity and mortality.
- The dosing of acyclovir in obese patients is challenging as it is not clear which type of body weight to use in calculating the dose.
- There is a lack of consistency in measuring the dose as using the actual body weight could result in acyclovir toxicity and utilizing the ideal body weight could lead to low ineffective concentrations.

Findings:

- This review summarizes the literature on the available evidence of acyclovir dosing in obese patients.
- Herein we point out the inconsistency of acyclovir dosing in obese patients and the potential subsequent events that occur from either higher or lower doses.
- As a result, we proposed some recommendations based on the published evidence for acyclovir dosing considering body weight and kidney function.

The recommended dose of acyclovir in severe infections such as herpetic encephalitis is 10mg/kg intravenous (IV) every 8 hours in normal kidney function to be given for 14–21 days.[13-15] Adequate acyclovir dosing is challenging as evidence varied on which body weight to use in calculating the dose. However, more attention should be given to the acyclovir dose to avoid either subtherapeutic drug concentration or overdose toxicity, which may worsen the patient's clinical outcomes. It has been recommended to use the ideal body weight (IBW) or adjusted body weight to calculate the acyclovir dose in obese patients.[13,15] It is not clear; however, which body weight should be utilized in obese patients. Using the IBW to calculate the acyclovir dose in obese patients may result in subtherapeutic concentrations,[16] while the use of actual body weight (ABW) may result in acyclovir neurotoxicity and/or nephrotoxicity.

Objectives

The objective of this scoping review is to explore existing evidence and identify the knowledge gap regarding acyclovir dosing for obese patients afflicted with herpes encephalitis and subsequently formulate a recommendation for the optimal acyclovir dosage within this specific patient population. Herein we propose a potential strategy that could assist clinicians in prescribing acyclovir doses to minimize the risk of suboptimal therapeutic levels or toxic overdose.

Methods

Search strategy

This review adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) guideline.[17] A comprehensive literature review was conducted with no language limitations. The databases searched on 26 May 2023 to include articles (from inception to 26 May 2023) were MEDLINE (1966 to 26 May 2023), EMBASE (1947 to 26 May 2023), Scopus (2004 to 26 May 2023), Web of Science (1960 to 26 May 2023), and CINAHL (1982 to 26 May 2023). The appropriate keywords were selected with the help of a medical librarian. The following keywords were used to ensure that relevant articles will be retrieved in the study: (acyclovir OR aciclovir OR zovirax OR acycloguanosine) AND (*"herpes simplex"* OR "herpes zoster" OR herpe* OR shingles OR chickenpox OR hsv* OR *varicella* OR *zoster* OR *"simplex vir*"*) *AND* (*"body weight"* OR *"BMI"* OR *"body mass"* OR *"body composition*"* OR *obes** OR *"overweight"* OR *"overweight"* OR *"body fat"* OR *"waist"*).

Study screening and selection

The retrieved articles were imported to Covidence software (www.covidence.org), a web-based software for conducting review screening to complete the process of screening and selecting the relevant articles that will be included in the study. The process of screening was done independently by 2 reviewers. Title and abstract screening was conducted, followed by a full-text review, to select the relevant articles that investigated acyclovir dosing based on body weight. Studies that were conducted on children or neonates or animals were excluded. Also, studies in different languages that could not be translated and articles with limited data availability were excluded. The data collected were study design, sample size, the population studied, age, sex, diagnoses, any reported adverse effects attributed to acyclovir, main study findings, and the limitations of the studies. We tried to contact the authors of the conference abstracts and we received a reply from 1 author with a supporting document.[18]

Results

The database search yielded 893 studies, of which 382 duplicate studies were removed (Figure 1). Following the screening of titles and abstracts, 41 articles underwent a thorough full-text review. Additionally, 3 articles were identified through the screening of the bibliographies of the included studies and were subsequently added to the review. Furthermore, through our ongoing surveillance of acyclovir articles, we identified 1 recent study published after the databases search was performed and was added to our review.[19] Following the full-text review, 19 articles were excluded, resulting 22 articles included in our scoping review. Study design varied across studies: 3 prospective studies, 8 retrospective studies, 10 case reports, and 1 pharmacist survey. Of these studies, 6 were conference abstracts.

Table 1 summarizes the articles included in this scoping review. Two of the prospective studies compared the pharmacokinetics of acyclovir between obese and nonobese

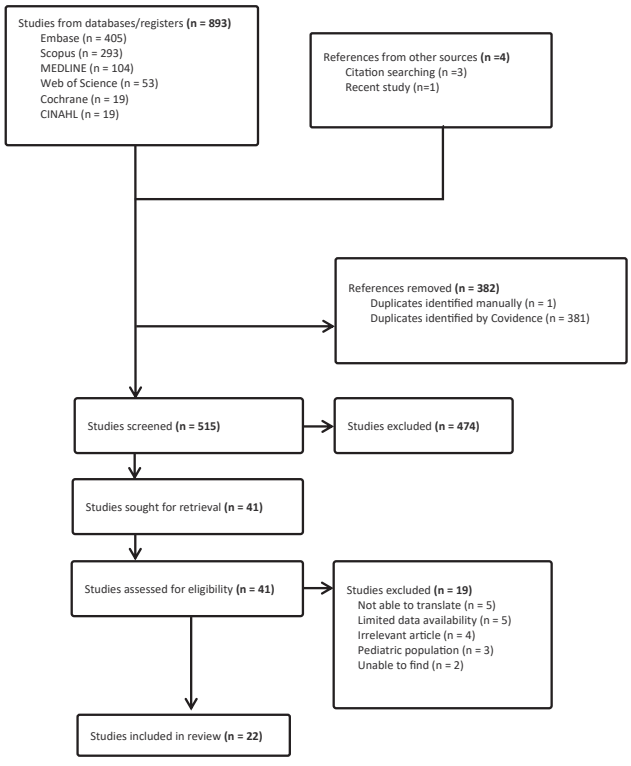


Figure 1. Flow chart of the literature search of studies in acyclovir dosing in herpes encephalitis.

individuals after single dose of 5 mg/kg.[16,20] The third prospective study compared the trough and peak acyclovir and its metabolite concentrations in 2 patient groups classified based on their kidney function.[21] Two retrospective studies observed the risk factors associated with acyclovir adverse effects between obese and nonobese patients.[18,22] Several other retrospective studies have reported the prevalence of acyclovir-induced adverse effects and the risk factors associated with them.[19,23-26] A retrospective single center study developed a statistical decisional tree based on the favorable and unfavorable outcomes to determine the dose of acyclovir based on body weight.[27] Most of the case reports were obese patients who developed acyclovir-induced nephrotoxicity and/or neurotoxicity when they received an acyclovir dose based on their ABW.[12,28,30,33,35] Also, a report of high dose of acyclovir 15mg/kg resulted in nephrotoxicity in 2 patients.[34] The remaining case reports discussed acyclovir treatment in immunocompromised or dialysis patients or drug interaction with acyclovir resulting in nephrotoxicity.[29,31,36]

Discussion

Acyclovir dosing in encephalitis and body weight

Acyclovir is the drug of choice in the treatment of herpes encephalitis as reported by previous randomized controlled trials.[4,37] The dosing of acyclovir is dependent upon body weight as nonobese patients receive the dose based on their ABW; however, the most appropriate weight for calculating the dose in obese patients remains unclear. Obesity is defined by World Health Organization by having a BMI of $\geq$ 30 kg/m^2.[38] Some studies suggest the use of the IBW while others recommend the use of adjusted body weight.[16,39] The IBW is defined based on the sex of the individual through the following equations:

$$\text{Males IBW} = 50.0\ \text{kg} + 2.3\ \text{kg} \times (\text{patient height in inches, above 60 inches})$$

$$\text{Females IBW} = 45.5\ \text{kg} + 2.3\ \text{kg} \times (\text{patient height in inches, above 60 inches}).[40]$$

The adjusted body weight is IBW plus 40% the difference between ABW and IBW and calculated using the following equation:

$$\text{Adjusted body weight} = \text{IBW} + (0.4 \times (\text{ABW} - \text{IBW})).[39]$$

A conference abstract in 1991 recommended that the dose of acyclovir should be calculated based on the IBW of obese patients. The study found that there was no difference between the volume of distribution of acyclovir in obese and nonobese patients when the dose was given based on ABW (42.5 $\pm$ 8.7 vs. 43.5 $\pm$ 5.7 L/1.73 m^2) which results in lower volume of distribution/kg in obese individuals and hence results in higher concentrations if similar doses were given. However, the use of ABW to measure the acyclovir dose in obese patients resulted in higher acyclovir maximum concentrations (7.45$\pm$1026 vs. 14.91 $\pm$7.24 mg/L, $P = 0.0025$), and higher concentration of acyclovir after 12 hours of completion of infusion (0.06$\pm$ 0.04 vs. 0.13 $\pm$0.04 mg/L, $P = 0.014$).[20] It is clear that the use of ABW to calculate acyclovir dose resulted in acyclovir-induced adverse effects.[12,28,30,33,35]

Acyclovir adverse effects

Acyclovir is generally a well-tolerated drug; however, it may cause mild gastrointestinal symptoms (nausea and vomiting), nephrotoxicity, and neurotoxicity.[41] In most cases, a maximum concentration above 25 mg/L may result in moderate to severe adverse effects. Moderate adverse effects interfere with daily routine or treatment, while severe adverse effects disrupt daily routine or require urgent management.[42] Attention should be given to the nephrotoxicity and neurotoxicity associated with acyclovir, as they can rapidly deteriorate a patient's condition and result in unfavorable outcomes.[43] Acyclovir induced nephrotoxicity is manifested by acute kidney injury (AKI).[44] The neurotoxicity of acyclovir tends to be reversible and the manifestations include a combination of disorientation, confusion, hallucination, tremors, ataxia, myoclonic jerks, slurred speech, seizures, or other motor or psychological disturbances.[45]

Most of the studies reported that reversible nephrotoxicity of acyclovir is common. The prevalence of acyclovir associated nephrotoxicity has been reported to range from 13%[24] to 21%[25] and the onset of AKI is 4.3 $\pm$ 2.5 days following the start of the acyclovir treatment.[46] Literature comparing the prevalence of AKI among obese and nonobese is scarce. It has been reported that the prevalence of AKI in obese patients was 8.5%[19] and 29%[25] as opposed to 13% in nonobese patients.[25] This variability in reported prevalence could be attributed to the small

Table 1
Summary of the studies included in the current review[a]

Author, year	Study design	Sample size	Population	Age (y)	Sex Female n(%)	Diagnosis	Adverse effects of acyclovir	Acyclovir dose	Main results	Limitations of the study
Davis et al, 1991[20]	A prospective study. (Conference abstract)	12	5 nonobese, 7 morbidly obese healthy individuals	Not reported	12 (100)	Not reported	Not reported	5 mg/kg IV	Mean pharmacokinetic parameters either adjusted for body weight or not adjusted were not different between the 2 groups. Doses based on TBW of obese resulted in higher concentration than nonobese C_{max} 7.45±1026 vs 14.91 ±7.24 mg/L, $P = 0.0025$ C12h 0.06± 0.04 vs 0.13 ±0.04 mg/L, $P = 0.014$ Recommend using IBW to calculate acyclovir dose in morbidly obese patients.	• Conference abstract with limited data availability.
Turner et al, 2016[16]	A prospective controlled study	14	The mean body mass index of 45.0 (obese) and 22.5 (nonobese) kg/m^2 (7 obese, 7 nonobese)	≥ 18 Obese: 54.3 ± 9.6 Nonobese: 53.0 ± 16.3	6 (85.7)	Patients receiving acyclovir as a prophylaxis in case of anticipated chemotherapy induced neutropenia	Not reported	A single dose of IV acyclovir, 5 mg/kg, infused over 60 min using TBW (non obese) and IBW (obese)	The use of IBW for dose calculation of IV acyclovir in obese patients leads to lower acyclovir serum concentration than dosing by TBW in non-obese patients. The maximum concentration and the area under the curve of obese patients were lower than those of nonobese patients (15.2 mg*h/L ± 2.9 vs 24.0 mg*h/L ± 9.4; $P = 0.011$), (5.8 ± 0.9 vs. 8.2 ± 1.3 mg/L; $P = 0.031$) respectively.	• Low acyclovir dose • Acyclovir was received as a prophylaxis and not a treatment • The study included only obese patients with BMIs of ≥ 40 kg/m^2 and didn't study the obese patients with BMIs less than 40 kg/m^2
Kacirova et al, 2022[21]	A prospective study	26	Patients receiving IV acyclovir between June 2019 and October 2021 Weight 78 ± 16 kg, height 169 ± 11 cm. patients were divided into 2 groups based on the kidney function	70 ± 20	17 (65.4)	Serious infection caused by HSV or VZV	One patient developed reversible neurotoxicity	Acyclovir doses were adjusted based on renal function.	Trough concentrations of acyclovir ranged from 0.8 to 7.6 mg/L and its metabolite CMMG ranged from 0.12 to 2.30 mg/L. The peak acyclovir concentrations and its metabolite ranged from 6.3 to 25.7 mg/L and 0.47 to 2.70 mg/L respectively. The ratio of the trough concentrations ranged from 0.07 to 0.63 and the ratio of the peak concentrations ranged from 0.03 to 0.24. For patients with renal dysfunction, even if they received lower doses adjusted to their reduced renal function, their trough acyclovir and its metabolite plasma concentrations were significantly higher than other patients.	• The reference range of acyclovir could not be determined because of variations in data and side effects between patients.
Li et al, 2016[22]	A retrospective cohort study (conference abstract)	94	Critically ill patients who received a high dose of IV acyclovir from January 2015-March 2016 in a tertiary university hospital. BMI; Obese 32% ≥ 30kg/m^2 vs	Not reported	Not reported	Not reported	AKI 21%	(10mg/kg/dose) IV acyclovir based either on AdjuBW, TBW or IBW.	46% of obese patients were dosed using their AdjuBW and 32% were dosed based on their TBW. There was no difference in the rates of AKI between the obese (23%) and lean patients (20%) ($P = 0.79$)	Retrospective nature of the study

			lean<30kg/m^2 The patients were grouped based on the type of body weight used to calculate the dose.						There was no association between body weight used and the development of AKI in both obese and lean patients ($P = 0.38$).	
Twilla et al, 2016[23]	A retrospective analysis (conference abstract)	468	Patients ≥ 18 year old and received IV acyclovir for ≥ 48 h. Patients admitted to Healthcare adult hospitals receiving IV acyclovir over nearly 5 year period.	≥ 18	Not reported	Not reported	AKI	344 patients dosed on TBW, 73 on IBW, and 51 on adjusted body weight	The weight used to calculate the dose of acyclovir didn't affect AKI occurrence. Increasing BMI was suggested as a risk factor for AKI. 24% of patients dosed on TBW developed AKI, 27% of patients dosed on IBW developed AKI and 31% of patients dosed on AdjuBW developed AKI.	• No statistical analysis is present for the results. • Missing data.
Ryan, 2018[24]	A single-center retrospective cohort study of prospectively acquired electronic data.	268	Patients in infectious diseases, neurology and neurointensive care departments who received parenteral acyclovir over the period of (2010–2016)	≥16	AKI 19 (54) Non AKI 112 (48)	Indications for acyclovir therapy were: Encephalitis (19%), Herpes zoster (18%), Meningitis (12%), disseminated herpes simplex (4%), Others (47%).	AKI prevalence is 13%.	Doses ranged from 5 mg/kg to 15 mg/kg 3 times daily, resulting in a total daily dose of 235 to 3600 mg	The study compared different risk factors with acyclovir AKI and found that the daily dose of acyclovir and the previous chronic kidney disease were the main risk factors associated with AKI.	They weren't able to get the patient's height from the records so either using IBW or TBW in correlation with the BMI was not studied.
Barber et al, 2019[25]	A retrospective case-control study	115	Patients who received IV acyclovir for >48 h at a university-affiliated tertiary hospital and referral center over a period of 4-y	≥18 The median age [IQR] was 54 (37–64) y	60 (52.5)	62% had indications for CNS infection with 15% had positive microbiological results.	Acyclovir-associated nephrotoxicity (21%).	Median dose of 10.1 (9.9–11.4) mg/kg and 8.1 (6.0–10.2) mg/kg based upon IBW and TBW, respectively	One-fifth of the patients developed nephrotoxicity. Obesity ($P = 0.04$) and vancomycin (OR, 4.73; 95% CI, 1.57–14.25) were the risk factors associated with nephrotoxicity.	• The study did not assess the improvement and other clinical outcomes associated with acyclovir dose.
Mulvey et al, 2020[26]	A retrospective observational chart review (conference abstract)	135	Adult patients admitted to a medical center with a body mass index greater than or equal to 30 kg/m^2 who received at least 48 h of high-dose IV acyclovir therapy during the study period of January 2014 to August 2019 51 (efficacy analysis) 84 (safety analysis)	Not reported	Not reported	Not reported	Nephrotoxicity occurred in 22.2%, 19.2%, and 22.7% of patients in the IBW, Adjusted body weight, and TBW groups respectively with no statistical significant difference ($P = 1$).	Patients were classified based on IBW, TBW or AdjuBW used to calculate the acyclovir dose.	Treatment failure occurred in 3 out of 51 patients (1 patient in the IBW group, 2 patients in the AdjuBW group, $P = 0.445$). There was no significant difference in the median length of stay ($P = 0.977$) or median duration of IV therapy ($P = 0.78$). The use of AdjuBW resulted in a lower acyclovir dose and lower exposure to acyclovir.	• Limited sample size. • Limited data availability. • Retrospective nature of the study.
Mulatero et al, 2022[27]	A descriptive, retrospective, monocentric study	76	Patients hospitalized in intensive care and conventional hospitalization sector. They classified the patients based on their weight (≥79 kg or less than 79 kg) regardless of acyclovir dose	≥16	39 (51%)	Herpetic encephalitis.	Poor outcomes occurred in 35 (51%) of patients.	(10–15 mg/kg every 8 h) for 14 to 21 d, with differences between patients.	The mortality rate was 12%, 49% had full recovery, and 39% had long term sequalae. The risk factors for poor outcomes were persistence of confusion, aphasia, and impaired consciousness lasting more than 5 d, superinfection, status epilepticus, and the length of stay in the intensive care unit. A statistical decision tree was developed to provide a treatment management plan based on the patient body weight and acyclovir dose with the suggestion of	• The retrospective nature of the study. • The decision tree limitation in relation to variation as it predicts the results based on data which may result in over fitting and high variance.

(*continued on next page*)

Table 1 (*continued*)

Author, year	Study design	Sample size	Population	Age (y)	Sex Female n(%)	Diagnosis	Adverse effects of acyclovir	Acyclovir dose	Main results	Limitations of the study
									increasing the dose of low weight patients(< 79kg)	
Carter et al, 2023[18]	A retrospective, multicenter, observational cohort study (conference abstract)	15	Five patients were obese and 10 were not obese	≥18 Obese 48 (41–69) Nonobese 53 (63–66)	Not reported	HSV encephalitis.	AKI did not differ significantly between groups (20% in obese group vs. 10% in nonobese group, $P = 1.00$)	Two patients in the non-obese group (20%) received IV acyclovir dosed on adjusted body weight. Obese patients received the dose based on IBW.	ICU admission, mechanical ventilation, and vasopressor support occurred more frequently among patients with unfavorable outcomes as compared to those with favorable outcomes.	• Limited by the retrospective nature of the study.
Zelnicek et al, 2023[19]	A retrospective cohort study	94	Adult obese patients with a BMI ≥ 30 kg/m^2 prescribed IV acyclovir ≥ 48 h between 1 January 2014 and 31 August 2021 at an academic medical center. The patients were divided into 2 groups based on the type of weight used to calculate acyclovir dose: AdjuBW group and IBW group.	≥18 Mean(SD) 54 (16.1)	52 (55.3)	Meningitis/ encephalitis (58.5%). Disseminated infection (14.9%). Ocular/retinal necrosis (10.6%).	AKI (8.5%)	In the total population: 650 [550–800] mg/dose IBW group: 600 [500–700] mg/dose Adjusted body weight group: 800 [700–850] mg/dose	Patients in the adjusted body weight group received higher doses of acyclovir compared to IBW group $P \leq 0.0001$. No patients developed AKI in the adjusted body weight group while 13.1% in the IBW group developed AKI.	• Retrospective nature of the study.
Hernandez et al, 2009[28]	Case-report	1	Obese patient. (108.9kg)	60	0	Herpes encephalitis.	On day 3, increased serum creatinine (SCr) 2.8 g/dL and blood urea nitrogen (BUN) 32 mg/dL, (nephrotoxicity).	1 g IV over 60 min every 8 hours. (10mg/kg every 8hrs) TBW.	The patient received an acyclovir dose based on TBW then he developed renal failure (SCr and BUN concentrations peaked at 5.5 g/dL and 57 mg/dL) which was reversible upon discontinuation of acyclovir. Hemodialysis was not feasible due to the patient status.	• Subjected to case-report limitations such as lack of generalizability, potential biases, absence of control groups, limited data availability, confounding factors, absence of statistical analysis, and lower evidence strength.
Klein et al, 2010[29]	Case-report	1	Female patient	53	1	Varicella zoster encephalitis.	Not reported	10 mg/kg IV every 8 h	Confusion (viral meningio-encephalitis slowly improved over 2 d and all symptoms resolved in 1 week.	• Subjected to case report bias.
Sacchetti et al, 2014[30]	Case-report	1	Morbidly obese female. (147 kg)	69	1	Zoster shingles. Followed by neurotoxicity which may be a result of 1 of the following: herpes zoster encephalitis vs. uremic encephalitis vs. acyclovir neurotoxicity.	Acute confusion and word-finding difficulties, occasional visual hallucinations.	One dose of 1.5 g IV acyclovir (10 mg/kg) given based on TBW and discharged home on oral acyclovir.	Her creatinine clearance raised to 7.4 mg/dl which is due to acyclovir nephrotoxicity . The patient developed neurological symptoms which were resolved upon discontinuation of acyclovir.	• Subjected to case report bias.
Berry et al, 2014[12]	Case-report	1	Obese female	75	1	Confusion, ataxia, and visual hallucinations with previously assumed herpetic stomatitis. Suspicion of herpes encephalitis.	AKI and neurotoxicity.	Oral acyclovir, 200 mg 5 times a day then switched to (IV) acyclovir at 10 mg/kg every 8 h. Dose based on TBW	High levels of acyclovir and its metabolite were detected in the CSF which led to confirmation of acyclovir neurotoxicity. Several factors should be considered when giving acyclovir to patients. Herpes viruses not detected within the CSF and the mouth swab so	• Subjected to case report bias.

									discontinuation of acyclovir and the neurological symptoms were related to acyclovir toxicity.	
Kageyama et al, 2016[31]	Case-report	1	Male- dialysis patient with chronic renal failure	75	0	Herpes zoster infection.	Neurotoxicity.	IV administration of 125 mg acyclovir 2.1 mg/kg per day, dose adjusted based on patient medical condition.	His serum acyclovir concentrations after the last dose before discontinuation were 5.96 mg/L at 15 h, 4.73 mg/L at 43 h and 1.49 mg/L at 63 h after the last acyclovir which is correlated with his neurological symptoms. Improvement after acyclovir discontinuation.	• Subjected to case report bias. • Close monitoring should be given to dialysis patients to avoid acyclovir toxicity.
Smith et al, 2016[32]	Case-report	1	Obese (BMI) of 40.4 kg/m^2 [height, 162.6 cm; (IBW), 54.7 kg; (TBW), 106.7 kg; body surface area, 2.10 m^2	61	1	Meningitis and chorioretinitis due to varicella-zoster virus.	The patient didn't develop acyclovir adverse effects.	1000 mg (9.4 mg/kg) IV every 8 h (q8h)	The Cmax and C8h concentrations for the obese patient far exceeded those reported for non-obese healthy volunteers receiving doses of 10 mg/kg q8h, contributing to decreased renal function in that obese patient. It is more important to consider renal function rather than weight-based dosing of acyclovir.	• Limited pharmacokinetics sampling time. • Single case report with not enough evidence.
Ferrari et al, 2016[33]	Case-report (conference abstract)	1	Obese (112 kg, BMI 33.8 kg/m^2)	35	0	Facial nerve paralysis, diplopia, and ptosis. Suspicion of Ramsay Hunt Syndrome which is a complication of varicella-zoster virus.	AKI	10 mg/kg every 8 h per day (3300 mg) dose based on TBW.	His serum creatinine increased from 0.84 to 3.55 mg/dl. The birefringent crystals in the urine suggested the association between acyclovir and renal toxicity. The reduction of acyclovir dose resulted in improvement and the patient retained normal renal function. Careful monitoring of renal function in acyclovir dosing is suggested.	• Subjected to case report bias.
Martinot et al, 2019[34]	Case-reports	2	Male and female (81 and 80, 81kg)	Not reported	1	Herpes encephalitis	AKI	Started 10mg/kg q8h Then increased to 15mg/kg q8h upon the confirmation of varicella-zoster infection.	Elevated creatinine levels of 209 and 459 μmol/l as a result of acyclovir nephrotoxicity. Clinical improvement after acyclovir discontinuation.	• Subjected to case report bias.
Sallevelt et al, 2020[35]	Case-report	1	Obese female (165 cm, 95 kg, BMI 35 kg/m^2)	71	1	Herpes zoster ophthalmicus and meningoencephalitis	AKI and neurotoxicity	10 mg/kg q8h IV Based on TBW.	Acyclovir peak plasma concentration reached 41.3 mg/L. Multiple hemodialysis sessions resulted in the relief of nephrotoxicity and neurotoxicity.	• Subjected to case report bias.
Berger et al, 2023[36]	Case-report	2	Patients with renal transplant	69 and 49	0	Herpes encephalitis OR herpes zoster infection (VZV) infection	Nephrotoxicity and/or neurotoxicity due to acyclovir interaction with MFF	High dose acyclovir 10 mg/kg twice daily 1000 mg valacyclovir orally twice daily.	Case A, and B acyclovir trough concentrations reached 3.5 mg/L and 3.2 mg/L respectively due to drug interaction with MFF. Case A developed neurotoxicity Improvement after acyclovir dose reduction or discontinuation.	• Subjected to case report bias.

Abbreviations used: AKI, acute kidney injury; BMI, body mass index; BUN, blood urea nitrogen; C8h, the plasma concentration at 8 h of infusion completion; C12h, concentration after 12 h of infusion completion; Cmax, maximum concentration; CSF, cerebrospinal fluid; HSV, herpes zoster virus; IBW, ideal body weight; ICU, intensive care unit; IQR, interquartile range; IV, intravenous; MFF, mycophenolate mofetil; OR, odd ratios; SCr, serum creatinine; TBW, total body weight; VZV, varicella zoster virus.

[a] 21 studies are included in the table and the pharmacist survey is included in the text.

sample size of the included studies, impeding accurate identification of acyclovir induced AKI prevalence. It is important to consider that the body weight used to dose acyclovir can significantly impact the prevalence percentage observed in obese patients. As acyclovir is primarily eliminated through the kidneys (Appendix 1), elevated concentrations of acyclovir could potentially lead to crystal formation in the kidneys, contributing to the development of AKI. The patients could experience decreased urine output, elevated blood urea nitrogen and serum creatinine levels, nausea, and vomiting. The risk factors associated with acyclovir nephrotoxicity are obesity and concomitant administration of nephrotoxic drugs.[25] Because AKI is prevalent among patients receiving acyclovir, using the IBW and kidney function to calculate the dose of acyclovir is of clinical importance.[24] They recommend monitoring the serum creatinine in the first few days after starting acyclovir treatment. Barber et al suggested that obesity was the only comorbidity associated with acyclovir nephrotoxicity in patients taking high acyclovir dose of 10 mg/kg.[25] As a result more attention should be given to obese patients in case of acyclovir dosing especially if it is given with other nephrotoxic drugs such as vancomycin. They used RIFLE (Risk, Injury, Failure, Loss of kidney function, and End-stage kidney disease) classification to stratify acyclovir nephrotoxicity. On the other hand, Twilla et al reported that obesity is not a risk factor for acyclovir-induced AKI, however, an increased BMI category increased the number of patients who developed AKI.[23] A retrospective study supported the idea that obesity did not increase the chance of developing AKI regardless of the body weight used to calculate the acyclovir dose. However, this study could be subjected to bias as there are no detailed patient characteristics.[22] In obese patients, there is not enough evidence of acyclovir pharmacokinetics. A case of an obese patient who suffered from kidney dysfunction, his maximum concentration and the concentration after 8 hours of infusion completion far exceed those of nonobese patients even after dose correction based on kidney function. However, acyclovir clearance and volume of distribution were similar in obese and nonobese patients. That patient was fortunate to not develop acyclovir adverse effects, but there should be concerns to not give the dose based on ABW.[20,32] These results should be supported by large scale studies to be able to assess the pharmacokinetics of acyclovir in obese patients.

The prevalence of neurotoxicity attributed to acyclovir was poorly reported in literature but is thought to be of rare occurrence.[45] Either acyclovir overdose or decreased acyclovir clearance could lead to acyclovir and its metabolite accumulation and hence neurotoxicity. The mean onset of acyclovir induced neurotoxicity symptoms is 3.1 days ± 4.3 (0.2−28) after the start of acyclovir treatment.[45] Although neurotoxicity associated with acyclovir is relatively rare, it is crucial to remain vigilant for its possibility when prescribing the medication and to closely monitor patients for any signs during the course of treatment. The CMMG level in the plasma and the cerebrospinal fluid will be a good indication to differentiate between acyclovir-induced neurotoxicity and other causes of neurological symptoms. If CMMG level is high it indicates that neurotoxicity is most likely acyclovir induced and the patient should receive the most appropriate management to mitigate the neurotoxicity such as acyclovir dose reduction, discontinuation, or hemodialysis.[21,47] The ratio of CMMG/acyclovir serum concentrations were higher in patients with renal impairment. So measuring the concentration of acyclovir as well as its metabolite is important for optimizing the therapeutic outcomes.[21] Some studies have developed methods for analyzing CMMG using LC-MS.[48] Therefore, if laboratories are equipped with the necessary instruments, it could potentially be analyzed. However, it is not currently conducted as a routine measurement. Some patients required hemodialysis to reduce acyclovir concentrations.[35,49,50] Attention should be given to kidney impairment as it is a main factor that could lead to acyclovir induced neurotoxicity, especially in dialysis patients.[5,51]

Acyclovir therapy failure

Given that herpes encephalitis is a potentially fatal infection, it is imperative to ensure that acyclovir reaches therapeutic concentrations. It has been suggested that the target trough acyclovir concentration should exceed the 50% inhibitory concentration, which is 0.56 mg/L for herpes simplex virus and 1.125 mg/L for varicella-zoster virus.[52] It has been reported that using IBW resulted in lower serum concentrations of acyclovir in obese patients compared to normal patients.[16] Evidence reporting consequences of reduced acyclovir exposure in encephalitis is lacking. Further studies are needed.

Evidence summary

A prospective cohort study compared the acyclovir plasma concentration in obese and nonobese patients after a single dose of 5 mg/kg every 8 hours concluded that using IBW to dose acyclovir in obese patients may result in lower concentration and hence lower therapeutic effect.[16] However, it's important to note that ABW was utilized to calculate the dose for nonobese patients, whereas IBW was used for dosing obese patients. The area under the curve and the maximum concentration of acyclovir in obese patients were 15.2± 2.9 mg*h/L and 5.8 ± 0.9 mg/L, respectively), which is lower than those of nonobese patients (24.0 ± 9.4 mg*h/L; *P* value = 0.011 and 8.2 ± 1.3 mg/L; *P* value = 0.031), respectively. As a result, the authors recommended the use of adjusted body weight to calculate the acyclovir dose in obese patients. However, the dose of 5 mg/kg every 8 hours is lower than recommended for encephalitis and further investigations utilizing different higher doses would best describe the acyclovir plasma concentrations in obesity.

Patients with renal impairment receive reduced acyclovir doses; however, despite receiving low doses they are still prone to have high acyclovir concentrations.[21] A few retrospective studies reported that the type of body weight used to calculate acyclovir dose in obese patients did not increase the prevalence of nephrotoxicity,[22,23] the length of stay, and the duration of IV therapy.[26] A recent retrospective study conducted by Zelnicek et al, compared the acyclovir dose and the prevalence of nephrotoxicity among obese patients and found that those who received it based on their adjusted body weight received higher doses and had no toxicity, while those who received it based on their IBW received lower doses and had 13.1% nephrotoxicity. This discrepancy could be contributed to small sample size and patients variability.[19] Barber et al, suggested that obesity is a risk factor for developing acyclovir induced nephrotoxicity,[25] which was not supported by Carter et al.[18] Among the other risk factors that could predispose the

Table 2
An approach to calculate the appropriate acyclovir dose in the management of herpes encephalitis[a]

BMI	Normal kidney function (CrCl >50 ml/min/1.73 m^2) (3)	Decreased kidney function (CrCl ≤50 ml/min/1.73 m^2) (60)	Augmented kidney function (CrCl >130 ml/min/1.73 m^2) (46)	Comment
Nonobese (BMI <30 kg/m^2)	10 mg/kg/dose every 8 h (ABW)	If CrCl: 25–50 ml/min/1.73 m^2 10 mg/kg/dose every 12 h 10–25 ml/min/1.73 m^2 10 mg/kg/dose every 24 h <10 ml/min/1.73 m^2 5 mg/kg/dose every 24 h	10–15 mg/kg/dose every 8 h	Therapeutic drug monitoring is recommended in specific patients.[b] Measure acyclovir concentration at steady state (after ~24h of consistent dosing) 1-If acyclovir maximum concentration after the infusion < 25 mg/L, so continue current regimen. 2- If the acyclovir maximum concentration is >25 mg/L so decrease the dose or if necessary stop acyclovir if signs or symptoms of nephrotoxicity or neurotoxicity. 3- If acyclovir trough concentration immediately before the next dose is <1.125 mg/L for varicella viruses and < 0.56 mg/L for herpes simlex viruses so increase the dose gradually to obtain desired therapeutic concentration. 4- If acyclovir trough concentration immediately before the next dose is ≥1.125 mg/L for varicella viruses and ≥ 0.56 mg/L for herpes simlex viruses so continue the current regimen. Monitor the urine output. Monitor any signs or symptoms of nephrotoxicity and neurotoxicity and intervene if necessary.
Obese (BMI ≥30 kg/m^2)	10 mg/kg/dose every 8 h (AdjuBW)	If CrCl: 25–50 ml/min/1.73 m^2 10 mg/kg/dose every 12 h (AdjuBW) 10–25 ml/min/1.73 m^2 10 mg/kg/dose every 24 h (AdjuBW) <10 ml/min/1.73 m^2 5 mg/kg/dose every 24 h (AdjuBW)	10–15 mg/kg every 8 h (AdjuBW)	

Abbreviations used: ABW, actual body weight; AdjuBW, adjusted body weight; BMI, body mass index; CrCl, creatinine clearance.
[a] The level of evidence of these recommendations is low.
[b] Therapeutic drug monitoring is recommended in patients with impaired kidney function (CrCl <50 ml/min/1.73 m^2), obese patients and not improving encephalitis patients while on acyclovir treatment.

patients to acyclovir induced nephrotoxicity were the dose, presence of chronic kidney disease, and concomitant use of nephrotoxic drugs.[24,25]

On the other hand, a few case reports described obese patients who received 10 mg/kg of acyclovir based on their ABW to treat herpes encephalitis.[12,28,30,33,35] They reported that these patients developed either nephrotoxicity or neurotoxicity or both as a result of acyclovir's high blood concentration around 41 mg/L.[12,35] For instance, 1 of these case reports was acyclovir-induced neuropathy, when the patient received acyclovir to treat shingles he developed neurological symptoms, which may be either due to acyclovir supratherapeutic concentration or herpes encephalitis symptoms. By considering these case reports it is clear that the inconsistent way of acyclovir dosing between different practices may potentially lead to poor clinical outcomes.[30]

A pharmacist survey of dosing practices of acyclovir based on body weight in herpes encephalitis was emailed to a group of licensed health care professionals. Out of 264 respondents, 240 completed the survey and they were mostly clinical pharmacists. There was a wide variation in the acyclovir dose calculation between obese and nonobese patients. When the patient ABW is 109% of his IBW, 57.4% of pharmacists chose to use the IBW to calculate the acyclovir dose. But if the patient ABW is 137% of his IBW, 57% of the pharmacists chose to use IBW to measure the acyclovir dose. Similarly, if the patient ABW is 157% IBW, 49.4% of them chose the IBW to calculate the dose, but 46.2% indicated that they would use the adjusted body weight. On the other hand, for the patients with highest body weight (ABW is 178% of IBW), most of the pharmacists (49.8%) selected to use the adjusted body weight to calculate the acyclovir dose, while 43.4% would use the IBW. The results of this survey indicate that pharmacists tend to calculate the dose of acyclovir based on adjusted body weight especially when the patient's weight increases. However, there was a clear lack of consistency on which body weight to use in the calculation of acyclovir dose, especially in overweight patients.[39]

Is it only the weight that could affect acyclovir exposure?

There is an important factor that was neglected and more focus should be given to it, augmented renal clearance (ARC). ARC is a condition that may result in lower concentrations of renal eliminated medications which may affect the patient's clinical outcomes. ARC accounts for around 14% to 80% and is more common among critically ill patients.[53] ARC is defined when the glomerular filtration rate is greater than 130 mL/min/1.73 m^2. As acyclovir is mainly eliminated through the kidney, the plasma concentrations of acyclovir are markedly decreased in the case of ARC which may result in subtherapeutic concentrations in serious infections such as life-threatening herpes encephalitis. Higher doses of acyclovir are usually recommended in case of ARC.[52]

Based on the available evidence Table 2 presents the recommended acyclovir doses in herpes encephalitis considering the 2 main factors that affect acyclovir concentrations are the type of bodyweight and kidney function. It is suggested that obese patients receive their acyclovir dose based on the adjusted body weight.[16] Moreover, we suggest that obese patients who suffer renal impairment to monitor their acyclovir concentration. Dose adjustment is required if the creatinine clearance (CrCl) is less than or equal 50 ml/min/1.73 m^2. If the CrCl is from 25–50 ml/min/1.73 m^2 so the patient will receive the normal dose but to be given every 12 hours, and if the CrCl is from 10–25 ml/min/1.73 m^2 so the patient will receive the normal dose but only once daily. And if the CrCl is less than 10 ml/min/1.73 m^2 so the patient will receive one-half the normal dose once daily.[54] In patients with ARC the patients could receive up to the maximum recommended manufacturer dose with therapeutic drug monitoring to avoid toxic concentrations.

The limitations of this study is the limited interventional studies on what type of body weight to be used to calculate acyclovir dose will result in the best plasma concentrations. Moreover, most of the studies have small sample sizes which could not address the actual pharmacokinetics in obese patients. The large number of case reports and conference abstracts in our study, however, will be important in case of rare diseases such as herpes encephalitis, they may be biased. As a result, the suggested approach is based on low quality evidence due to scarcity of interventional studies.

Conclusion

There is insufficient evidence to recommend a specific body weight for calculating acyclovir dosage in obese patients. A tailored approach is necessary to administer acyclovir doses appropriately, considering both body weight and renal function, whether it's augmented or decreased. One point of interest that there may be an association between the subtherapeutic concentration of acyclovir in obese patients and the phenomena of ARC, which requires further investigation. Therefore, it is recommended to conduct additional studies comparing the dosing regimens and concentrations of IV acyclovir in both obese and nonobese patients. These studies should aim to correlate the resulting concentrations with the patient's kidney function and efficacy and adverse effects of acyclovir, including nephrotoxicity and neurotoxicity.

Disclosure

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References

1. Levitz RE. Herpes simplex encephalitis: a review. *Heart Lung*. 1998;27(3):209–212.
2. Hjalmarsson A, Blomqvist P, Skoldenberg B. Herpes simplex encephalitis in Sweden, 1990-2001: incidence, morbidity, and mortality. *Clin Infect Dis*. 2007;45(7):875–880.
3. Tunkel AR, Glaser CA, Bloch KC, et al. The management of encephalitis: clinical practice guidelines by the Infectious Diseases Society of America. *Clin Infect Dis*. 2008;47(3):303–327.
4. Whitley RJ, Alford CA, Hirsch MS. Vidarabine versus acyclovir therapy in herpes simplex encephalitis. *N Engl J Med*. 1986;314(3):144–149.
5. Bradshaw MJ, Venkatesan A. Herpes simplex virus-1 encephalitis in adults: Pathophysiology, diagnosis, and management. *Neurotherapeutics*. 2016;13(3):493–508.

6. Hartley L, Evans J, MacDermott NE. Management and outcome in viral meningo-encephalitis. *Paediatr Child Health*. 2011;21(11):488–494.
7. Schaeffer HJ, Gurwara S, Vince R, Bittner S. Novel substrate of adenosine deaminase. *J Med Chem*. 1971;14(4):367–369.
8. Schaeffer HJ, Beauchamp L, de Miranda P, Elion GB, Bauer DJ, Collins P. 9-(2-hydroxyethoxymethyl) guanine activity against viruses of the herpes group. *Nature*. 1978;272(5654):583–585.
9. van der Meer JW, Versteeg J. Acyclovir in severe herpes virus infections. *Am J Med*. 1982;73(1A):271–274.
10. Hellden A, Odar-Cederlof I, Larsson K, Fehrman-Ekholm I, Linden T. Death delusion. *BMJ*. 2007;335(7633):1305.
11. Hellden A, Odar-Cederlof I, Diener P, et al. High serum concentrations of the acyclovir main metabolite 9-carboxymethoxymethylguanine in renal failure patients with acyclovir-related neuropsychiatric side effects: an observational study. *Nephrol Dial Transplant*. 2003;18(6):1135–1141.
12. Berry L, Venkatesan P. Aciclovir-induced neurotoxicity: Utility of CSF and serum CMMG levels in diagnosis. *J Clin Virol*. 2014;61(4):608–610.
13. Acyclovir (Systemic) (Lexi-Drugs) - Lexicomp. Available at: https://online.lexi.com/lco/action/doc/retrieve/docid/patch_f/1763800?cesid=1GYbi0RBBcN&searchUrl=%2Flco%2Faction%2Fsearch%3Fq%3Dacyclovir%26t%3Dname%26acs%3Dtrue%26acq%3Dacyclo. Accessed January 23, 2024.
14. Workowski KA, Bachmann LH, Chan PA, et al. Sexually transmitted infections treatment guidelines, 2021. *MMWR Recomm Rep*. 2021;70(4):1–187.
15. Taylor M, Gerriets V. Acyclovir. Available at:. In: *StatPearls*. StatPearls Publishing; 2024. http://www.ncbi.nlm.nih.gov/books/NBK542180/. Accessed February 6, 2024.
16. Turner RB, Cumpston A, Sweet M, et al. Prospective, controlled study of acyclovir pharmacokinetics in obese patients. *Antimicrob Agents Chemother*. 2016;60(3):1830–1833.
17. Tricco AC, Lillie E, Zarin W, et al. PRISMA extension for scoping reviews (PRISMA-ScR): Checklist and explanation. *Ann Intern Med*. 2018;169(7):467–473.
18. Carter C, Crotty M, Sylvain K, Nolen R, Welch E, Tieu J. Impact of obesity on clinical outcomes in patients receiving acyclovir for hsv encephalitis. *Crit Care Med*. 2023;51(1):184 (a slide deck presented at the SCCM congress has been provided by the authors).
19. Zelnicek TD, Siegrist EA, Miller JL, Neely SB, Higuita NIA, White BP. Comparison of dosing strategies in obese patients prescribed intravenous acyclovir and evaluation of rate of acute kidney injury. *Int J Antimicrob Agents*. 2023;62(2):106871.
20. Davis RL, Quenzer RW, Weller S, Blum R. *Acyclovir pharmacokinetics in morbid obesity, abstract 765. In 31st Interscience Conference on Antimicrobial Agents and Chemotherapy. American Society for Microbiology, Washington DC, USA*. 1991.
21. Kacirova I, Urinovska R, Sagan J. Therapeutic monitoring of serum concentrations of acyclovir and its metabolite 9-(carboxymethoxymethyl) guanine in routine clinical practice. *Biomed Pharmacother*. 2022;156:113852.
22. Li J, Durkin J. Evaluation of Iv acyclovir dosing and nephrotoxicity in critically ill obese and lean patients. *Crit Care Med*. 2016;44(12):400.
23. Twilla J, Broyles J, Sanchez A, Finch CK. An evaluation of acyclovir dosing in obesity. *Pharmacotherapy*. 2016;36(12):E208–E209.
24. Ryan L, Heed A, Foster J, Valappil M, Schmid ML, Duncan CJA. Acute kidney injury (AKI) associated with intravenous aciclovir in adults: incidence and risk factors in clinical practice. *Int J Infect Dis*. 2018;74:97–99 ((Ryan, Schmid, Duncan) Department of Infection and Tropical Medicine, Newcastle upon Tyne Hospitals NHS Foundation Trust, Newcastle upon Tyne, United Kingdom).
25. Barber KE, Wagner JL, Stover KR. Impact of obesity on acyclovir-induced nephrotoxicity. *Open Forum Infect Dis*. 2019;6(4):ofz121.
26. Mulvey N, Jain S, Falsetta K, Doan TL. Assessing the clinical impact of intravenous acyclovir dosing in obese patients: should we be using ideal, adjusted, or total body weight? *Open Forum Infect Dis*. 2020;7(Suppl 1):S102–S103.
27. Mulatero M, Boucekine M, Felician O, et al. Herpetic encephalitis: which treatment for which body weight? *J Neurol*. 2022;269(7):3625–3635.
28. Hernandez JO, Norstrom J, Wysock G. Acyclovir-induced renal failure in an obese patient. *Am J Health Syst Pharm*. 2009;66(14):1288–1291.
29. Klein NC, McDermott B, Cunha BA. Varicella-zoster virus meningoencephalitis in an immunocompetent patient without a rash. *Scand J Infect Dis*. 2010;42(8):631–633.
30. Sacchetti D, Alawadhi A, Albakour M, Rapose A. Herpes zoster encephalopathy or acyclovir neurotoxicity: a management dilemma. *BMJ Case Rep*. 2014;2014:bcr2013201941 ((Sacchetti, Alawadhi) Department of Internal Medicine, Saint Vincent Hospital, Worcester, MA, United States).
31. Kageyama R, Hashizume H. Neurotoxicity induced by the recommended acyclovir dosing in a dialysis patient with herpes zoster: a case letter. *J Dermatol*. 2016;43(3):339–340.
32. Smith TC, Kim JH, Gast CM, Benefield RJ. Pharmacokinetics of acyclovir in a morbidly obese patient with renal impairment. *Int J Antimicrob Agents*. 2016;47(4):340–341.
33. Ferrari V, Nicolini E, Dentali F, et al. Acute renal failure in a obese patient on acyclovir treatment. *Ital J Med*. 2016;10(Supplement 2):43.
34. Martinot M, Klein A, Demesmay K, et al. Acute renal failure related to high doses of acyclovir (15 mg/kg/8 h) during treatment of varicella zoster virus encephalitis. *Antivir Ther*. 2019;24(4):309–312.
35. Sallevelt BT, Smeijsters EH, Egberts TC, van der Elst KC, Mudrikova T. Acute renal and neurotoxicity due to weight-based dosing of intravenous acyclovir: how to dose in obese patients? *Clin Infect Pract*. 2020;7-8:100046 ((Sallevelt, Smeijsters, Egberts, van der Elst) Department of Clinical Pharmacy, University Medical Center Utrecht, Utrecht, Netherlands).
36. Berger FA, Nagtegaal JE, Malingre MM. Two cases of a pharmacokinetic interaction between (val)acyclovir and mycophenolate mofetil. *Clin Infect Pract*. 2023;17:100211 ((Berger, Nagtegaal, Malingre) Department of Clinical Pharmacy, Meander Medical Centre, Amersfoort, Netherlands).
37. Skoldenberg B, Alestig K, Burman L, et al. Acyclovir versus vidarabine in herpes-simplex encephalitis - randomized Multicenter study in Consecutive Swedish patients. *Lancet*. 1984;2(8405):707–711.
38. World Health Organization(WHO). Obesity and overweight. Available at: https://www.who.int/news-room/fact-sheets/detail/obesity-and-overweight. Accessed August 20, 2023.
39. Wong A, Pickering AJ, Potoski BA. Dosing practices of intravenous acyclovir for herpes encephalitis in obesity: results of a pharmacist survey. *J Pharm Pract*. 2017;30(3):324–328.
40. Devine BJ. Gentamicin therapy. *Drug Intell Clin Pharm*. 1974;8(11):650–655.
41. Balfour HH. Acyclovir therapy for herpes-zoster - advantages and adverse-effects. *JAMA*. 1986;255(3):387–388.
42. Bean B, Aeppli D. Adverse effects of high-dose intravenous acyclovir in ambulatory patients with acute herpes zoster. *J Infect Dis*. 1985;151(2):362–365.
43. Wei YP, Yao LY, Wu YY, et al. Critical review of synthesis, toxicology and detection of acyclovir. *Molecules*. 2021;26(21):6566.
44. Pierson-Marchandise M, Gras V, Moragny J, et al. The drugs that mostly frequently induce acute kidney injury: a case – noncase study of a pharmacovigilance database. *Br J Clin Pharmacol*. 2017;83(6):1341–1349.
45. Brandariz-Nunez D, Correas-Sanahuja M, Maya-Gallego S, Martin Herranz I. Neurotoxicity associated with acyclovir and valacyclovir: a systematic review of cases. *J Clin Pharm Ther*. 2021;46(4):918–926.
46. Yalcinkaya R, Oz FN, Kaman A, et al. Factors associated with acyclovir nephrotoxicity in children: data from 472 pediatric patients from the last 10 years. *Eur J Pediatr*. 2021;180(8):2521–2527.
47. Kacířová I, Uřinovská R, Sagan J. The importance of therapeutic drug monitoring in acyclovir dosage optimization. *Klin Farmakol Farm*. 2023;36(4):129–135.
48. Determination of acyclovir and its metabolite 9-carboxymethoxymethylguanide in human serum by ultra-high-performance liquid chromatography-tandem mass spectrometry - Urinovska - 2021 - Journal of Separation Science - Wiley Online Library. Available at: https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/epdf/10.1002/jssc.202100241. Accessed January 23, 2024.
49. Leikin JB, Shicker L, Orlowski J, Blair AT, McAllister K. Hemodialysis removal of acyclovir. *Vet Hum Toxicol*. 1995;37(3):233–234.
50. Krieble BF, Rudy DW, Glick MR, Clayman MD. Acyclovir neurotoxicity and nephrotoxicity-the role for hemodialysis. *Am J Med Sci*. 1993;305(1):36–39.
51. Adair JC, Gold M, Bond RE. Acyclovir neurotoxicity: clinical experience and review of the literature. *South Med J*. 1994;87(12):1227–1231.
52. Abdalla S, Briand C, Oualha M, et al. Population pharmacokinetics of intravenous and oral Acyclovir and oral Valacyclovir in pediatric population to optimize dosing regimens. *Antimicrob Agents Chemother*. 2020;64:e01426-20 ((Abdalla, Oualha, Beranger, Benaboud, Treluyer, Zheng, Foissac, Gana, Boujaafar, Bouazza, Hirt) EA7323, Evaluation des therapeutiques et pharmacologie perinatale et pediatrique, Universite de Paris, Paris, France).
53. Mahmoud SH, Shen C. Augmented renal clearance in critical illness: an important consideration in drug dosing. *Pharmaceutics*. 2017;9(3):36.
54. Blum MR, Liao SH, de Miranda P. Overview of acyclovir pharmacokinetic disposition in adults and children. *Am J Med*. 1982;73(1A):186–192.

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Appendix 1.

Acyclovir pharmacokinetics

Absorption

The oral absorption of acyclovir is slow and incomplete, resulting in low bioavailability (15% to 30%),[1,2] therefore it is mainly used as an IV injection. When treating herpes encephalitis usually the desired acyclovir concentrations are above those obtained through an oral regimen.

Distribution

Acyclovir has a volume of distribution at steady state of 48.0 ± 13.6 L/1.73 m^2 (22.5–101 L/1.73 m^2) and low plasma protein binding (22% to 33%).[3,4] Therefore, acyclovir is more distributed in the lean mass of the body than fat tissues. The body composition of normal weight patients consists of 80% lean mass to 20% fat mass, in contrast overweight patients have higher fat mass of 40%. As a result if the acyclovir is to be given based on the ABW it will result in higher concentrations in overweight patients.[5] If the obese patients received the acyclovir dose based on their ABW there is a high chance to develop acyclovir adverse effects.[6]

The plasma concentration-time curve of acyclovir follows a 2 compartments open model with zero-order infusion and first order elimination kinetics.[7] Following IV administration, the plasma concentration-time curve has 2 phases, a rapid short phase of distribution between central and peripheral compartments followed by linear first order elimination kinetics phase.[8] It has been recommended that the target trough acyclovir concentration to be above the 50% inhibitory concentration which is 0.56 mg/L and 1.125 mg/L for HSV and varicella zoster viruses, respectively.[9] The maximum plasma concentration of acyclovir ranged between 4.52 ± 0.31 and 22.7 ± 10.4 mg/L for single IV doses of 2.5–15 mg/kg/dose.[10] A multiday, multidoses study found the first dose peak plasma concentration of 4 different IV acyclovir doses of 2.5, 5.0, 10.0, and 15.0 mg/kg to be 5.06 ± 2.70, 8.15 ± 1.24, 14.99 ± 3.63, and 21.32 ± 6.76 mg/L, respectively. And the steady state plasma concentration of these 4 doses were 6.78 ± 2.84, 9.73 ± 0.97, 15.06 ± 1.24, and 20.64 ± 2.50 mg/L, respectively. The linear correlation between the steady state concentration and the area under the curve with the dose indicates the dose-independent pharmacokinetics of acyclovir.[7]

Metabolism and excretion

Acyclovir is mainly eliminated by the kidney and, therefore, requires dosage adjustment in case of kidney dysfunction.[3] The average renal clearance of acyclovir is 260 ± 81 mL/min/1.73 m^2 and it has a half-life of 3–5 hours.[4,7] The renal clearance of acyclovir is 3 times higher than that of the creatinine which indicates that acyclovir is eliminated through glomerular filtration and tubular excretion.[3] Acyclovir clearance after IV infusion or rapid injection is independent of the dose.[1,11] A very small portion of acyclovir is subjected to hepatic metabolism by the aldehyde oxidase enzyme and is converted to a renally eliminated compound (CMMG). The percentages of acyclovir and its metabolite CMMG found in the urine of healthy volunteers were 68 ± 14% and 2.4% to 13.6%, respectively.[7] The half-life and plasma concentrations of acyclovir were significantly increased in patients with end-stage renal disease even with receiving the recommended modified dose.[7,12] Acyclovir doesn't compete with creatinine for tubular excretion, so elevated creatinine level during acyclovir therapy is a real indication of nephrotoxicity.[13,14] The mechanisms beyond acyclovir-induced nephrotoxicity are the deposition of acyclovir needle shaped crystals within the renal tubules of the kidneys, tubular damage or dysfunction that occurs independent of crystal deposition, and acute tubular necrosis.[15] Pre-existing renal dysfunction or dehydration may worsen acyclovir-induced nephrotoxicity.[4] Rapid injection of acyclovir is also associated with AKI so the recommendation is to infuse acyclovir over 1 hour to avoid acyclovir nephrotoxicity.[1,16] Suggested maneuvers to prevent acyclovir-induced nephrotoxicity is to routinely monitor the kidney function, ensure the slow infusion of acyclovir and provide adequate hydration to maintain a urine output of 75 ml/h.[17]

References

1. de Miranda P, Blum MR. Pharmacokinetics of acyclovir after intravenous and oral administration. *J Antimicrob Chemother*. 1983;12(Suppl B):29–37.
2. Fletcher C, Bean B. Evaluation of oral acyclovir therapy. *Drug Intell Clin Pharm*. 1985;19(7-8):518–524.
3. de Miranda P, Whitley RJ, Blum MR, et al. Acyclovir kinetics after intravenous infusion. *Clin Pharmacol Ther*. 1979;26(6):718–728.
4. Gurgel Assis MS, Fernandes Pedrosa TC, de Moraes FS, et al. Novel Insights to enhance therapeutics with acyclovir in the management of herpes simplex encephalitis. *J Pharm Sci*. 2021;110(4):1557–1571.
5. Barras M, Legg A. Drug dosing in obese adults. *Aust Prescr*. 2017;40(5):189–193.
6. Sallevelt BT, Smeijsters EH, Egberts TC, van der Elst KC, Mudrikova T. Acute renal and neurotoxicity due to weight-based dosing of intravenous acyclovir: how to dose in obese patients? *Clin Infect Pract*. 2020;7-8:100046 ((Sallevelt, Smeijsters, Egberts, van der Elst) Department of Clinical Pharmacy, University Medical Center Utrecht, Utrecht, Netherlands).
7. Whitley RJ, Blum MR, Barton N, de Miranda P. Pharmacokinetics of acyclovir in humans following intravenous administration. A model for the development of parenteral antivirals. *Am J Med*. 1982;73(1A):165–171.
8. Metzler CM. Usefulness of the two-Compartment open model in pharmacokinetics. *J Am Stat Assoc*. 1971;66(333):49–53.
9. Abdalla S, Briand C, Oualha M, et al. Population Pharmacokinetics of intravenous and oral Acyclovir and oral Valacyclovir in pediatric population to optimize dosing regimens. *Antimicrob Agents Chemother*. 2020;64:e01426-20 ((Abdalla, Oualha, Beranger, Benaboud, Treluyer, Zheng, Foissac, Gana, Boujaafar, Bouazza, Hirt) EA7323, Evaluation des therapeutiques et pharmacologie perinatale et pediatrique, Universite de Paris, Paris, France).
10. Laskin OL, Longstreth JA, Saral R, de Miranda P, Keeney R, Lietman PS. Pharmacokinetics and tolerance of acyclovir, a new anti-herpesvirus agent, in humans. *Antimicrob Agents Chemother*. 1982;21(3):393–398.
11. Brigden D, Bye A, Fowle AS, Rogers H. Human pharmacokinetics of acyclovir (an antiviral agent) following rapid intravenous injection. *J Antimicrob Chemother*. 1981;7(4):399–404.
12. Laskin OL, Longstreth JA, Whelton A, et al. Effect of renal failure on the pharmacokinetics of acyclovir. *Am J Med*. 1982;73(1A):197–201.
13. Ryan L, Heed A, Foster J, Valappil M, Schmid ML, Duncan CJA. Acute kidney injury (AKI) associated with intravenous aciclovir in adults: incidence and risk factors in clinical practice. *Int J Infect Dis*. 2018;74:97–99 ((Ryan, Schmid, Duncan) Department of Infection and Tropical Medicine, Newcastle upon Tyne Hospitals NHS Foundation Trust, Newcastle upon Tyne, United Kingdom).
14. Gunness P, Aleksa K, Koren G. The effect of acyclovir on the tubular secretion of creatinine in vitro. *J Transl Med*. 2010;8:139.
15. Sawyer MH, Webb DE, Balow JE, Straus SE. Acyclovir-induced renal failure. Clinical course and histology. *Am J Med*. 1988;84(6):1067–1071.
16. Bean B, Aeppli D. Adverse effects of high-dose intravenous acyclovir in ambulatory patients with acute herpes zoster. *J Infect Dis*. 1985;151(2):362–365.
17. Bradshaw MJ, Venkatesan A. Herpes Simplex Virus-1 encephalitis in adults: Pathophysiology, diagnosis, and management. *Neurotherapeutics*. 2016;13(3):493–508.